SAMARTH MUTALIKDESAI

• samarthmutalikdesai7@gmail.com • linkedin.com/in/samarth7 • Vijayapura, Karnataka • +91 8951371247

SUMMARY

Machine Learning enthusiast with hands-on experience in Python, Computer Vision, and Deep Learning. Completed a research internship at IIIT Dharwad focused on facial-part prediction and reconstruction. Experienced in AI applications using Python, OpenCV, MediaPipe, PyTorch, and scikit-learn, with a strong interest in AI/ML and research-oriented software development.

EXPERIENCE

Indian Institute of Information Technology Dharwad (IIIT Dharwad)- Intern

- Worked on a deep learning-based facial-part prediction and reconstruction problem.
- Explored facial landmark detection, image preprocessing, model training, and image reconstruction.
- Implemented experiments using Python, PyTorch, MediaPipe, and computer vision techniques.
- Evaluated reconstruction performance using MSE and SSIM.

Google Student Ambassador— Google

- Organized and conducted technology workshops and events on campus.
- Promoted Google technologies and programs among 200+ students and conducted 5+ workshops.
- Led a team to coordinate online and offline technology initiatives and AI-focused learning activities.

RESEARCH

Explainable AI-Based Facial Attention Monitoring System for Energy-Efficient Smart Classrooms — Research Paper
International Journal of Multidisciplinary Frontier Research (IJMFR)

- Developed an AI-based prototype for monitoring student facial attention in classroom environments.
- Applied Computer Vision and Explainable AI concepts using Python, OpenCV, MediaPipe, and Streamlit.
- Research work selected for publication.

PROJECTS

Smart Face Recognition Atte ndance System

- Built a real-time face recognition application using Python, OpenCV, and Streamlit.
- Implemented facial encoding and automated attendance logging with timestamps in CSV/Excel.

Student Score Predictio n

- Built a machine learning model using Python and scikit-learn to predict student performance.
- Visualized data using Matplotlib and evaluated performance using MAE and RMSE.

Smart Hostel Room Allocation & Roommate Compatib ility System

- Built a Django-based system using KNN for roommate compatibility and Decision Tree for room-type recommendation.
- Integrated student preferences, room capacity and study environment to support automated hostel room allocation.

EDUCATION

Bachelor of Computer Applications (BCA) **2024 – 2027**
A S Patil College of Commerce, Vijayapura, Karnataka CGPA: 8.4

TECHNICAL SKILLS

Programming: Python, C, C++, R, Data Structures
Machine Learning: Linear Regression, Model Evaluation, Feature Engineering, scikit-learn, CNN, KNN
Deep Learning: PyTorch, Neural Networks
Computer Vision: OpenCV, MediaPipe, Face Recognition
Libraries: NumPy, Pandas, Matplotlib,Django,
Tools: Google Colab, VS Code, Streamlit, Tableau

CERTIFICATIONS

- Deloitte Data Analytics Job Simulation — Data cleaning, analysis, and interpretation using real-world business scenarios.
- Simplilearn — Basics of Data Structures and Algorithms — Arrays, stacks, queues, linked lists and fundamentals.